

PREDICTIVE MODELING · REVIEW
WORKSHEET

Exercise worksheet: Predictive Models

From conceptual reasoning to model evaluation, with assertion-reason, multiple-choice, sum-of-statements, and open-ended challenges.

Maria Eduarda

30 questions

Commented answer key

Name: _____

Date: _____

How to use this worksheet

Solve it first without consulting the material. Then explain aloud why your answer makes sense. In data science, justifying a decision matters as much as selecting an option.

Assertion–reason convention

- A. Both statements are true, and the reason explains the assertion.
- B. Both are true, but the reason does not explain the assertion.
- C. The assertion is true, and the reason is false.
- D. The assertion is false, and the reason is true.
- E. Both statements are false.

PART 1

10 assertion–reason questions

Classify each pair using options A–E above.

1 Analyze the statements

Assertion: Feature scaling is especially important in KNN.

Reason: KNN compares examples by distance, and a variable on a much larger scale may dominate that calculation.

Answer: _____ **Brief justification:** _____

2 Analyze the statements

Assertion: Logistic Regression can produce a probability for the positive class.

Reason: The sigmoid function maps any real value to a number between 0 and 1.

Answer: _____ **Brief justification:** _____

3 Analyze the statements

Assertion: An accuracy of 95% always proves that a classifier is appropriate.

Reason: On imbalanced data, a model may ignore the rare class and still achieve high accuracy.

Answer: _____ **Brief justification:** _____

4 Analyze the statements

Assertion: A Random Forest tends to be more stable than one Decision Tree.

Reason: It combines votes from trees trained with variations in samples and features.

Answer: _____ **Brief justification:** _____

5 Analyze the statements

Assertion: Features should usually be scaled before K-Means.

Reason: The algorithm forms groups by minimizing distances to centroids.

Answer: _____ **Brief justification:** _____

6 Analyze the statements

Assertion: High precision means few false positives among positive predictions.

Reason: Precision is calculated as $TP \div (TP + FN)$.

Answer: _____ **Brief justification:** _____

7 Analyze the statements

Assertion: Recall matters when missing a positive case is costly.

Reason: Recall decreases when there are many false negatives.

Answer: _____ **Brief justification:** _____

8 Analyze the statements

Assertion: During training, KNN estimates coefficients to form an equation.

Reason: KNN stores examples and delays the decision until a new point arrives.

Answer: _____ **Brief justification:** _____

9 Analyze the statements

Assertion: Naive Bayes often works well for text classification.

Reason: Conditional independence simplifies how evidence from many words is combined.

Answer: _____ **Brief justification:** _____

10 Analyze the statements

Assertion: After cross-validation, a final test set is unnecessary.

Reason: Cross-validation estimates stability by repeating training and validation across folds.

Answer: _____ **Brief justification:** _____

PART 2

10 multiple-choice and sum questions

Select the best answer. In sum questions, add the values of all correct statements.

11 Problem type

Which option contains, in order, a classification problem and a regression problem?

- A) Predict house price; predict temperature.
- B) Detect fraud; estimate a purchase amount.
- C) Cluster customers; detect spam.
- D) Predict duration; segment images.

12 Metrics from the matrix

A model obtained $TP=36$, $TN=48$, $FP=12$, and $FN=4$. Which option is correct?

- A) Accuracy 84%, precision 75%, recall 90%.
- B) Accuracy 90%, precision 84%, recall 75%.
- C) Accuracy 75%, precision 90%, recall 84%.
- D) Accuracy 84%, precision 90%, recall 75%.

13 Choosing K in KNN

Which strategy is most appropriate?

- A) Always use $K=1$.
- B) Choose K using the test set.
- C) Test K values on validation data; for binary classification, an odd K can avoid ties.
- D) Use the number of features as K.

14 Elbow Method

In K-Means, the “elbow” indicates:

- A) Where increasing K stops reducing inertia substantially.
- B) The highest possible accuracy.
- C) The best Logistic Regression threshold.
- D) The number of missing values.

15 Data leakage

Which situation contains data leakage?

- A) Split train and test before fitting the scaler.
- B) Create a feature from information recorded after the event being predicted.
- C) Remove an identifier column.
- D) Calculate metrics on validation data.

16 Sum — evaluation

Add the values of all correct statements.

- 01 — Precision considers false positives.
- 02 — Recall considers false negatives.
- 04 — F1 combines precision and recall.
- 08 — R^2 can only range from 0 to 1.
- 16 — MAE uses the same unit as the target.

Sum: _____

17 Sum — clustering

Add the values of all correct statements.

01 — K-Means requires a value for K.

02 — A high Silhouette score suggests compact, separated groups.

04 — DBSCAN may label points as noise.

08 — Clustering requires known labels.

16 — Feature distance and scale influence the groups.

Sum: _____

18 Naive Bayes

Which sentence best describes the model?

A) It builds several trees and combines votes.

B) It updates probabilities with evidence, assuming conditional independence among features.

C) It forms groups using centroids.

D) It uses only the most distant example.

19 Metric aligned with risk

In screening for a serious disease, missing a sick person is costly. Which metric deserves special attention?

A) Recall for the sick class.

B) Accuracy only.

C) R^2 .

D) Inertia.

20 Feature engineering

To predict subscription churn, which feature is useful and safe?

A) "Churned next month."

B) A random customer identifier.

C) Days since last login, calculated only up to prediction time.

D) The final outcome copied into a new column.

PART 3

10 open-ended and practical questions

Show your reasoning. Diagrams, calculations, and pseudocode are welcome.

21 Build a confusion matrix

Among 100 transactions, 20 were fraudulent. The model detected 16 frauds and produced 10 false alarms. Build the TP/TN/FP/FN matrix, then calculate accuracy, precision, recall, and F1.

22 Explain and apply KNN

Explain KNN to a beginner. Then classify $Q=(5,4)$ with $K=3$ using $A=(4,4)$ Pink, $B=(6,5)$ Pink, $C=(5,2)$ Blue, $D=(9,8)$ Blue, and $E=(2,1)$ Blue. Compare distances, identify the three neighbors, predict the class, and explain why scaling matters.

23 One K-Means iteration

For values 1, 2, 3, 8, 9, and 10, use $K=2$ with initial centroids $C_1=2$ and $C_2=9$. Assign every value, calculate new centroids, and explain when the algorithm stops.

24 Regression versus classification

Explain the difference in output, examples, algorithms, and metrics. Classify apartment price, low/medium/high risk, quantity sold, and spam detection.

25 Feature engineering — Titanic

Using Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked, propose at least five new features or transformations. Explain each hypothesis and how you would prevent leakage.

26 Write code by hand — pipeline

Complete a pipeline that loads `/kaggle/input/competitions/titanic/train.csv`, inspects shape/types/missing values/target balance, splits with stratification, imputes, scales, one-hot encodes, trains Logistic Regression, and prints a confusion matrix and classification report.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
```

Complete the workflow below

27 Write code by hand — metrics

Without scikit-learn, write functions or pseudocode that receive TP, TN, FP, and FN and return accuracy, precision, recall, and F1. Handle division by zero.

28 Choose and defend a model

A company has a few thousand examples, needs explainable customer decisions, and has imbalanced classes. Compare Logistic Regression, KNN, Naive Bayes, Decision Tree, and Random Forest. Choose a starting point and describe validation.

29 Naive Bayes in words

Explain prior, likelihood, and posterior through a detective analogy. Why is the “naive” assumption unrealistic, and why can the model still work well for text?

30 Project challenge

Choose a real problem. Define the target, unit of observation, possible features, learning type, initial model, data split, main metric, baseline, and one ethical or operational risk.

ANSWER KEY

Answers and grading criteria

Check only after trying. Other well-justified open answers may also be valid.

Assertion–reason

1. A — distance depends on scale.

2. A — sigmoid maps scores to probabilities.

3. D — accuracy can hide minority-class failure.

4. A — diversity and voting reduce variance.

5. A — K-Means optimizes distances.

6. C — precision uses FP, not FN.

7. A — FN lowers recall.

8. D — KNN is instance-based.

9. A — the assumption simplifies computation.

10. D — a final independent test is still needed.

Multiple-choice

11. B

12. A

13. C

14. A

15. B

16. 23 (01+02+04+16)

17. 23 (01+02+04+16)

18. B

19. A

20. C

Open-answer guide

21. Matrix and metrics

TP=16, FN=4, FP=10, TN=70. Accuracy=86%; precision≈61.5%; recall=80%; F1≈69.6%.

22-24. Core concepts

KNN: nearest A, B, C → Pink by 2-1. K-Means: {1,2,3} → 2 and {8,9,10} → 9.
Regression predicts continuous values; classification predicts categories or class probabilities.

25-27. Features and code

Strong Titanic ideas include FamilySize, IsAlone, Title, FarePerPerson, HasCabin, Deck, ticket-group size, and age bands. Preprocessing must be fitted only on training data. Metric formulas must handle zero denominators.

28-30. Reasoning criteria

Reward coherent trade-offs, validation without test leakage, a metric aligned with error cost, features available at prediction time, a meaningful baseline, and ethical or operational risk analysis.